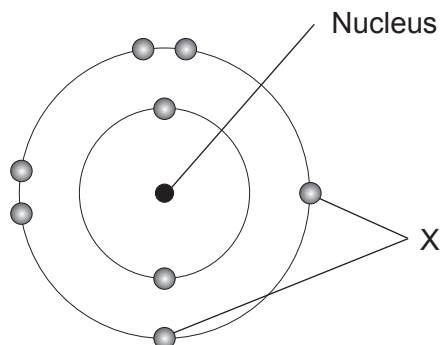


2. The Periodic Table is an arrangement of the chemical elements according to their atomic number, electron configuration and chemical properties.

An oxygen atom is shown below.



- (a) Complete the following sentences using **only** the words in the box. The words may be used once, more than once, or not at all.

protons	ions	electrons	neutrons
---------	------	-----------	----------

- (i) The nucleus of the oxygen atom contains and

.....

[2]

- (ii) The particles labelled X are called

[1]

- (iii) In a neutral atom the number of protons is equal to the number of

.....

[1]

- (b) Oxygen is found in Group 6 of the Periodic Table. Use the Periodic Table on page 20 to answer the following questions.

- (i) Give the electron configuration of oxygen.

[1]

.....

- (ii) State the name of **one** element in the same **group** as oxygen.

[1]

.....

- (iii) State the name of **one** element in the same **period** as oxygen.

[1]

.....



- (c) Water is a compound that contains one oxygen atom and two hydrogen atoms joined together.

- (i) Use the Periodic Table on page 20 to complete the following table. [2]

Element	Relative atomic mass
hydrogen	
oxygen	

- (ii) Use the information above to calculate the relative formula mass of H_2O . [1]

relative formula mass =

3445U101
05

10

